



Question Bank

Std. : 9 ★ CBSE

Sub. : Science

Biology Ch-2 Cell : The Building Block of Life (NCERT Solution)

Q.1 Where does a cell come from ?

Solution : A cell comes from a pre-existing cell through cell division. New cells are formed when existing cells grow and divide, ensuring continuity of life and transmission of genetic material from parent to daughter cells.

Q.2 How have technological interventions facilitated the creation of new knowledge in understanding the world beyond the naked eye ?

Solution : Technological tools like microscopes have enabled scientists to observe cells, organelles, and microorganisms beyond naked-eye limits. Advances such as electron microscopy and imaging techniques have revealed detailed structures and processes, leading to new knowledge about cell function, disease mechanisms, and life at the microscopic level.

Q.3 How is the cell structural and functional unit of life ?

Solution : A cell is the structural unit as all organisms are made of cells. It is the functional unit because all vital activities-metabolism, respiration, growth, and reproduction-occur within cells. Thus, cells form the building blocks and carry out essential life processes.

Q.4 How does a cell multiply ?

Solution : A cell multiplies by cell division. In this process, the parent cell grows, replicates its DNA, and divides into two daughter cells. This occurs through mitosis for growth and repair, ensuring each new cell receives identical genetic material.

Q.5 What argument would you give for the necessity of a cell wall in plants usually fixed in one place versus in animals usually moving from one place to the other ?

Solution : Plants remain fixed, so a rigid cell wall is necessary for support, protection, and maintaining shape against external forces. Animals are mobile and need flexibility; absence of a cell wall allows cells to change shape, enabling movement and specialized functions.

Q.6 What consequences would you predict for a plant cell if its cell wall were to become as flexible as a cell membrane ?

Solution : If the cell wall became as flexible as the cell membrane, the plant cell would lose rigidity and shape. It could not withstand turgor pressure, leading to swelling or bursting. The plant would lose structural support, become weak, and may collapse.

Q.7 Why is it important to cut the two potato pieces in roughly equal size and measure their initial weight before placing them in different liquids ?

Solution : Cutting equal-sized potato pieces and measuring initial weight ensures a fair test in osmosis experiments. It keeps surface area and mass constant, so any change in weight is due only to movement of water, not differences in size, giving accurate and comparable results.

Q.8 Do white flowers contain any pigment? Give reasons.

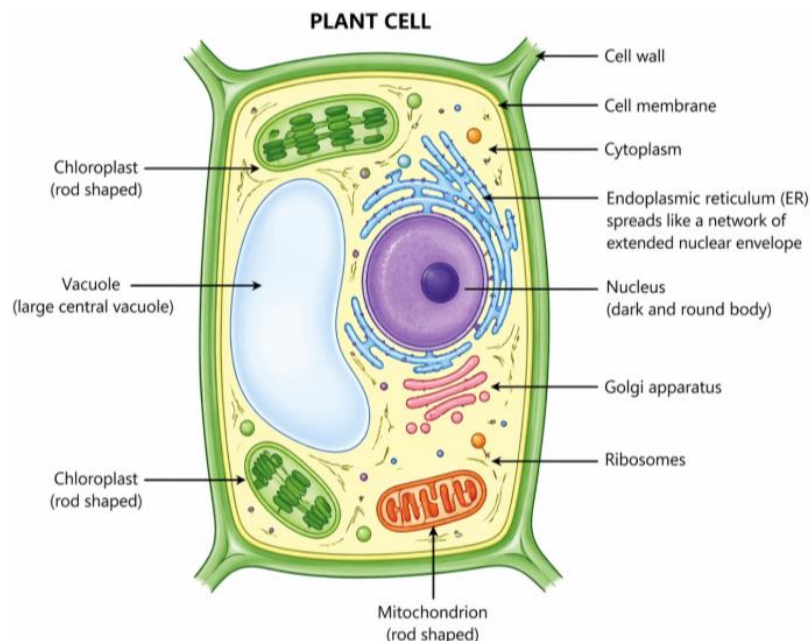
Solution : Yes, white flowers do contain pigments.

White flowers usually lack colored pigments like anthocyanins, but they often contain colorless or very faint pigments (like flavones). Also, their white appearance is mainly due to reflection of all wavelengths of light by air spaces in the petal cells, rather than complete absence of pigments.

Q.9 Draw a well-labelled schematic diagram of a plant or an animal cell using these clues-

1. Nucleus appears as a dark and round body inside the cell.
2. ER spreads like a network of extended nuclear envelope.
3. Mitochondria and chloroplasts are rod shaped.

Solution :



Q.10 Instead of many small ones, why does a cell not have a single giant mitochondrion? How does this relate to the concept of surface area ?

Solution : A cell does not have a single giant mitochondrion because it would be inefficient.

Reason (Surface Area concept): Many small mitochondria provide a larger total surface area compared to one big mitochondrion of the same volume. Since energy production (ATP formation) occurs on the inner membrane, more surface area → more energy production.

Also, smaller mitochondria allow :

- Better distribution of energy throughout the cell
- Faster exchange of materials

Q.11 If the skin cells start dividing by meiosis instead of mitosis, what do you think will happen to a cut on the skin ?

Solution : It would not heal properly.

Reason : Meiosis produces half the number of chromosomes and forms non-identical cells meant for reproduction. Skin repair needs identical diploid cells, which are produced by mitosis.

Q.12 Differentiate between the following pairs of terms based on the clues given in parentheses :

1. Cell membrane and cell wall (permeability)
2. RER and SER (structure)
3. Chloroplasts and chromoplasts (pigments)

Solution :

1. Cell membrane vs Cell wall (permeability)
 - Cell membrane: Selectively permeable (allows some substances to pass).
 - Cell wall: Freely permeable (allows most substances to pass).
2. RER vs SER (structure)
 - RER (Rough ER): Has ribosomes on surface; appears rough.
 - SER (Smooth ER): Lacks ribosomes; appears smooth.
3. Chloroplasts vs Chromoplasts (pigments)
 - Chloroplasts: Contain chlorophyll (green pigment).
 - Chromoplasts: Contain carotenoids (yellow, orange, red pigments).

Q.13 Two similar animal cells are placed in two different solutions :

- Cell X is placed in pure water.
- Cell Y is placed in a concentrated salt solution.

Cells are observed after some time. Cell X swells, and Cell Y shrinks. Which statement provides the correct explanation for the above observations?

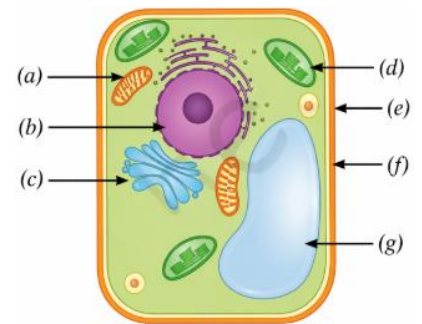
Options :

- (1) Salt molecules moved into Cell Y, causing it to shrink.
- (2) Water moved into Cell X and more water moved out of Cell Y than the salt solution entered in it.
- (3) Water moved into Cell X and moved out of Cell Y through the cell membrane. (4)
- (4) Solute movement caused osmosis in both cells.

Explanation : Osmosis = movement of water across a semipermeable membrane from dilute to concentrated solution → Cell X swells, Cell Y shrinks.

Q.14 Look at the diagram of a cell in figure. Identify the parts labelled from (a) to (g) and correctly match them with their functions given below :

1. Controlling all the activities of a cell.
2. Site of cellular respiration.
3. Storage organelle that also provides rigidity to the cell.
4. Separates the cell contents from surroundings.
5. Provides structural rigidity to the cell.
6. Packs and stores materials received from ER.
7. Helps in manufacturing food.

**Solution :**

1. → Mitochondrion → (ii) Site of cellular respiration
2. → Nucleus → (i) Controlling all the activities of a cell
3. → Golgi apparatus → (vi) Packs and stores materials received from ER
4. → Chloroplast → (vii) Helps in manufacturing food
5. → Cell membrane → (iv) Separates the cell contents from surroundings
6. → Cell wall → (v) Provides structural rigidity to the cell
7. → Vacuole → (iii) Storage organelle that also provides rigidity to the cell

Q.15 Which of the following option(s) of the pairs of cell organelles are correctly placed under the given categories ?

Option	Present in the plant cells	Absent in the animal cells
(i)	Leucoplast	Cell wall
(ii)	Mitochondria	Ribosome
(iii)	Cell wall	Golgi apparatus
(iv)	Lysosome	Endoplasmic reticulum

Options :

- (1) Pair (i) is correct. (2) Pair (ii) is correct.
(3) Pair (iii) is correct. (4) Pair (iv) is correct.

Explanation:

1. Leucoplast → present in plant cells; Cell wall → absent in animal cells
2. Mitochondria and ribosomes → present in both ✕
3. Golgi apparatus → present in both ✕
4. Lysosome and ER → present in both ✕

Q.16 Two students, Renu and Rohit, were having a discussion on the plastids. Renu emphasised that all parts of the plants, even roots, contain plastids. However, Rohit did not agree with the statement and told her that plastids are absent in plant roots since the roots are underground and do not need to perform photosynthesis. Who is correct? Justify your answer.

Solution : Renu is correct.

Justification : Plastids are present in all plant cells, including roots. In roots, plastids occur as leucoplasts (colorless plastids) which store food like starch. Although roots do not perform photosynthesis, they still contain plastids for storage and other functions.

Q.17 Mitochondria and chloroplasts are two important organelles in a plant cell. Discuss how these two organelles are structurally and functionally similar to each other, and different from each other.

Solution :

Similarities (structure + function) :

- Both are double-membrane organelles.
- Both have their own DNA and ribosomes (semi-autonomous).
- Both are involved in energy transformations in the cell.

Differences :**1. Mitochondria :**

- Inner membrane folded into cristae.
- Function: cellular respiration → releases energy (ATP).
- Present in both plant and animal cells.

2. Chloroplasts :

- Contain thylakoids (grana) with chlorophyll.
- Function: photosynthesis → makes food (glucose).
- Present only in plant cells.

Q.18 Which of the following pairs of cell organelles contains DNA?

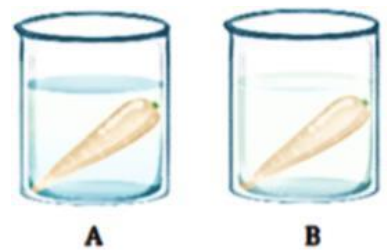
Options :

- (1) Chloroplasts, Ribosomes
- (2) Mitochondria, Nucleus
- (3) Golgi bodies, Ribosomes
- (4) Nucleus, Lysosomes

Explanation : Mitochondria and nucleus both contain DNA.

Q.19 A researcher carried out an experiment in which she took two carrots of similar size. She placed one carrot in plain water and the other carrot in concentrated salt solution. After 24 hours she recorded her observations.

1. What hypothesis does she want to test through this experiment?
2. What would you suggest for the improvement of this experiment?
3. Why does the carrot in plain water stay stiff and crunchy, but the carrot in concentrated salt solution become rubbery and limp?



Experimental set-up having carrot (a) in plain water, and (b) in salt solution

Solution :

1. **Hypothesis :** Osmosis causes water movement across cell membranes – cells gain water in a dilute solution and lose water in a concentrated solution.
2. **Improvement :**
 - Use carrot pieces of equal size and weight
 - Measure initial and final mass/length
 - Keep time and temperature constant
 - Repeat for accuracy
3. **Reason for observations :**
 - In plain water : water enters cells → cells become turgid → carrot stays stiff
 - In salt solution: water leaves cells → cells become flaccid/plasmolysed → carrot becomes limp

Q.20 Indicate the presence or absence of following structures in bacterial and animal cells :

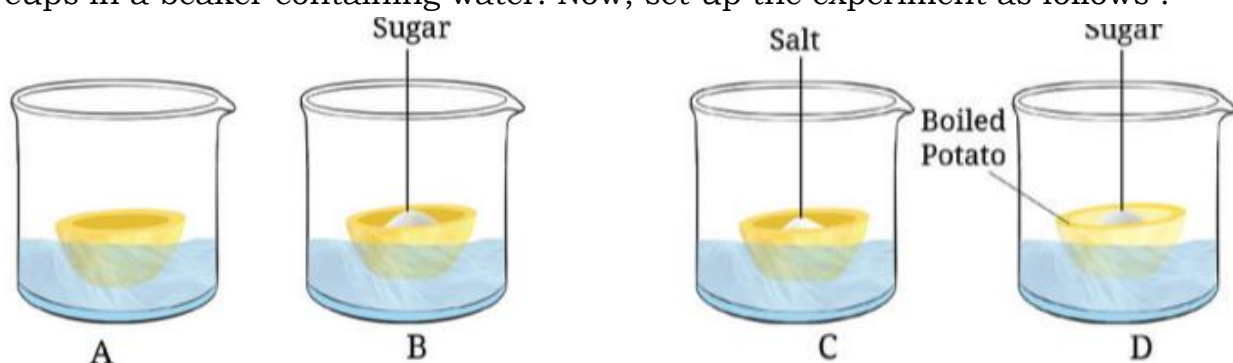
Structures in a cell	Bacterial cell	Animal cell
Chromosome		
Nucleus		
Mitochondria		
Golgi complex		
Chromoplasts		

Solution :

Structures in a cell	Bacterial cell	Animal cell
Chromosome	Present	Present
Nucleus	Absent	Present
Mitochondria	Absent	Present
Golgi complex	Absent	Present
Chromoplasts	Absent	Absent

Q.21 Carry out the following experiment :

Take four peeled potato halves and scoop each one out to make potato cups. One of these potato cups should be made from a boiled potato. Place each of the potato cups in a beaker containing water. Now, set up the experiment as follows :



1. Keep Cup A empty.
2. Add one teaspoon sugar in Cup B.
3. Add one teaspoon salt in Cup C.
4. Add one teaspoon sugar in the boiled potato in Cup D.

Observe the four potato cups at least two hours and answer the following questions :

1. Explain why water gathers in the hollowed portion of Cup B and Cup C.
2. Why is Cup A necessary for this experiment?
3. Explain why water does not gather in the hollowed portions of Cups A and D.

Solution :

1. Water gathers in Cups B and C due to osmosis. The sugar/salt solution inside is more concentrated, so water moves from the surrounding dilute solution (beaker water) into the cup through the potato tissue (semipermeable membrane).
2. Cup A acts as a control. It shows that without any solute, no water movement occurs into the cavity.
 - Cup A : No concentration difference → no osmosis → no water collects.
 - Cup D (boiled potato): Boiling destroys the semipermeable nature of cell membranes → osmosis does not occur → no water collects.

Q.22 Identify the pair that incorrectly matches the cell organelle with its function.

1. Ribosome – Protein synthesis
2. SER – Lipid and cellulose synthesis
3. Lysosome – Digestion of foreign agents

Solution : Incorrect pair : SER – Lipid and cellulose synthesis

Reason : SER synthesizes lipids, but cellulose is synthesized at the cell membrane, not by SER.

Q.23 What outcome do you expect, if all the mitochondria are removed from a eukaryotic cell ?

Solution : The cell would die.

Reason : Mitochondria are the main site of ATP (energy) production through cellular respiration. Without them, the cell cannot produce enough energy to carry out essential functions like transport, synthesis, and division, leading to cell death.

Q.24 Which phenomenon inhibits the formation of tumors in the human body? Can plants also develop tumors? Explain.

Solution : Phenomenon :

Contact inhibition – normal cells stop dividing when they come in contact with neighboring cells, preventing uncontrolled growth (tumor formation).

Plants and tumors :

Yes, plants can develop tumor-like growths (e.g., crown gall disease). However, these do not spread like cancer in animals because plant cells are fixed in place due to the cell wall, so they do not metastasize.

Q.25 The cell membrane of a cell is made up of proteins and lipids. Which cell organelles help in the synthesis of cell membrane? Write the path of these compounds from their site of synthesis to the cell membrane and show this through a labelled diagram.

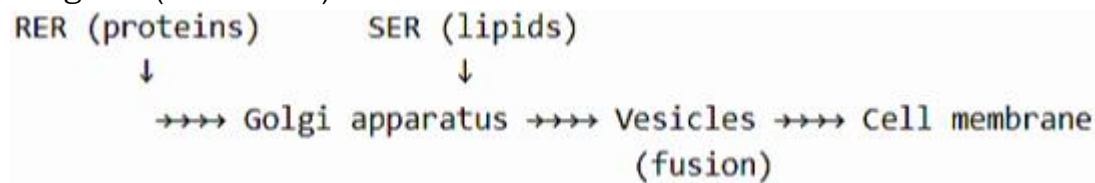
Solution : Organelles involved :

- RER (Rough Endoplasmic Reticulum) → synthesizes proteins
- SER (Smooth Endoplasmic Reticulum) → synthesizes lipids
- Golgi apparatus → modifies, packages, and transports them

Pathway :

Proteins (RER) + Lipids (SER) → transported to Golgi apparatus → modified & packed into vesicles → vesicles fuse with cell membrane

Diagram (schematic) :



Conclusion :

RER, SER, and Golgi together form and deliver components of the cell membrane.

Q.26 What would happen if gametes are formed by mitotic divisions ?

Solution : Gametes would be diploid (2n) instead of haploid (n).

Result :

- After fertilization → zygote becomes tetraploid (4n)
- Chromosome number would double every generation
- Leads to genetic imbalance and abnormal development

Conclusion :

Meiosis is essential to maintain a constant chromosome number across generations.

Q.27 A farmer, Deepa, was very happy with the harvest of amla (Indian Gooseberry) and lemons on her farm. However, she could sell only one-fourth of the produce in the local market. Recognising that a significant amount of produce may be lost post-harvest, she employed a traditional yet scientifically sound method to extend the shelf life of amla and lemons. She turned perishable produce into profitable products, such as pickles and sharbat. She used the excess produce to prepare pickles, murabbas, and sharbat by adding appropriate amounts of salt, sugar, or jaggery to small pieces of fruit and their juices. These were then stored in small glass bottles for sale, helping her prevent the wastage of post-harvest produce. This shift from farming to agro-processing would strengthen food security and boost the local economy, creating a sustainable model that cuts waste while increasing her income.

Based on the above passage answer the following questions :

1. Which scientific concept has the farmer applied in the preservation of the farm produce? (1)
2. How does the addition of high concentrations of salt and sugar create an environment that prevents the growth of spoilage-causing bacteria and fungi ? (1)
3. Suggest a healthy recipe of this kind for food preservation. (2)

OR

What are the scientific values addressed in this case? (2)

Solution :

1. Scientific concept: Osmosis (food preservation by high solute concentration).

2. How salt/sugar prevent spoilage :

High concentrations create a hypertonic environment, causing water to move out of microbial cells → plasmolysis → microbes cannot grow or survive.

3. Healthy recipe (example) :

Lemon pickle (low oil) :

- Cut lemons, add salt + turmeric + small amount of jaggery
- Keep in sunlight for a few days
- Store in clean glass jars

OR

Scientific values :

- Application of scientific knowledge
- Resource conservation (reducing wastage)
- Innovation and sustainability
- Economic efficiency

Q.28 What is the future of the development of synthetic cells using non-living chemicals? If a synthetic cell is developed, what may be the related ethical issues ?

Solution :

Future :

- Possibility of creating artificial (synthetic) cells from non-living chemicals
- Applications in medicine (targeted drug delivery, vaccines), industry (biofuels, enzymes), and research (understanding origin of life)
- Could enable custom-designed organisms for specific tasks

Ethical issues :

- Creation of life raises moral concerns
- Risk of misuse (biological weapons)
- Uncontrolled spread or ecological imbalance
- Questions about ownership/patenting of life forms
- Need for strict regulation and biosafety controls